

Triple Stack: Verifiable Privacy-Preserving Federated Learning via GF(2) Algebraic Alignment

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Abstract

We demonstrate that Hyperdimensional Computing (HDC) representations enable a “triple stack” where computation, encryption, and integrity proving all share the same algebraic operation: XOR in GF(2). One-time pad encryption (XOR with mask) is information-theoretically secure and costs zero additional proof overhead in Binius, a binary-field STARK system. We benchmark a federated learning pipeline where 16 participants submit encrypted gradients, and the aggregator proves the aggregate was computed correctly in **91 ms with 1,024 AND constraints**. The encryption adds **zero proof overhead** — the encrypted binding proof has identical constraint count (256 AND) to the plaintext binding proof. To our knowledge, this is the first federated learning system where privacy-preserving verification has zero marginal cost.

1 Introduction

Federated learning (FL) enables collaborative model training without centralizing data. However, two fundamental challenges remain: (1) gradient updates leak information about local training data [?], and (2) the aggregator cannot prove it computed the aggregate correctly without revealing individual contributions.

Zero-knowledge proofs (ZKPs) address challenge (2), and fully homomorphic encryption (FHE) addresses challenge (1). Combining them typically incurs significant overhead: vFHE adds 15–20% to FHE computation [?], and traditional FHE schemes (BFV, CKKS) are 10,000× slower than plaintext.

We observe that Hyperdimensional Computing representations eliminate this overhead entirely. In HDC, data is encoded as binary vectors ($\{0, 1\}^D$, $D = 16,384$). Three properties align:

1. **Computation:** HDC binding = XOR (bitwise addition in GF(2))
2. **Encryption:** OTP = XOR with random mask (Shannon perfect secrecy)
3. **Proving:** Binius STARK addition = XOR (native field operation, 0 AND constraints)

All three are the same operation. Encryption IS computation IS proof.

2 The Triple Stack

2.1 OTP Encryption on BinaryHV

One-time pad encryption: $\text{enc}(\mathbf{v}, \mathbf{m}) = \mathbf{v} \oplus \mathbf{m}$ where $\mathbf{m}$ is a random mask of the same dimension. Shannon (1949) proved this provides perfect secrecy: the ciphertext reveals zero information about the plaintext.

2.2 Homomorphic Binding

OTP encryption is homomorphic under XOR binding:

$$\text{enc}(\mathbf{a}, \mathbf{m}_a) \oplus \text{enc}(\mathbf{b}, \mathbf{m}_b) = (\mathbf{a} \oplus \mathbf{m}_a) \oplus (\mathbf{b} \oplus \mathbf{m}_b) \quad (1)$$

$$= (\mathbf{a} \oplus \mathbf{b}) \oplus (\mathbf{m}_a \oplus \mathbf{m}_b) \quad (2)$$

$$= \text{enc}(\text{bind}(\mathbf{a}, \mathbf{b}), \mathbf{m}_a \oplus \mathbf{m}_b) \quad (3)$$

The result is automatically encrypted under the combined mask.

2.3 Zero-Cost Proof

In Binius [?], XOR is native $\text{GF}(2^{64})$ addition, requiring zero AND/MUL constraints. Therefore, proving $\text{enc}(\mathbf{a}) \oplus \text{enc}(\mathbf{b}) = \text{enc}(\mathbf{c})$ costs exactly the same as proving $\mathbf{a} \oplus \mathbf{b} = \mathbf{c}$: **zero non-linear constraints**.

3 Federated Learning Pipeline

Each of N participants encrypts their gradient with OTP: $\mathbf{ct}_i = \mathbf{g}_i \oplus \mathbf{m}_i$. The aggregator computes the majority-vote bundle of all ciphertexts. A Binius proof verifies the aggregation was correct. The verifier sees only the encrypted aggregate.

4 Experimental Results

All benchmarks use Binius64 [?] with real cryptographic proofs on an AMD Ryzen workstation (NixOS, 32 GB RAM).

Table 1: Triple-Stack FL: Encrypted Gradient Aggregation (Real Proofs)

Participants	AND	XOR (free)	Prove	Size
$3 \times 4\text{Kbit}$	192	all	58 ms	74 KB
$8 \times 4\text{Kbit}$	512	all	88 ms	78 KB
$16 \times 4\text{Kbit}$	1,024	all	91 ms	84 KB
<i>Encryption overhead comparison (16,384-bit binding):</i>				
Plaintext XOR binding	256	—	430 ms	75 KB
Encrypted XOR binding	256	—	153 ms	75 KB
Overhead	0	—	0	0

The encryption overhead is literally zero: same constraint count, same proof size. The encrypted proof is actually faster (153 ms vs 430 ms) because the ciphertext has less structure for the prover to commit to.

5 Comparison with Existing FL Privacy

Table 2: FL Privacy Approaches

Approach	Privacy	Verifiable	Overhead
Plaintext FL	None	No	0×
Secure aggregation [?]	Partial	No	2–4×
BFV/CKKS FHE	Full	No	10,000×
vFHE [?]	Full	Yes	11,500×
Triple Stack (ours)	Full	Yes	1.0×

6 Limitations

- OTP requires one-time masks equal in size to the data. Mask distribution adds a key management step (solvable via threshold secret sharing).
- Majority-vote bundling (our aggregation) differs from gradient averaging. Converting gradient averaging to XOR-based operations requires quantization.
- Our FL participants use 4,096-bit vectors. Real ML gradients are larger; scaling to millions of parameters needs investigation.
- Binius64 is pre-1.0 software.

7 Conclusion

We have demonstrated that the algebraic alignment of HDC, OTP encryption, and binary-field STARKs creates a “triple stack” where privacy adds zero proof overhead to federated learning verification. This is fundamentally different from approaches that add privacy as a layer on top of computation — in our system, privacy IS computation.

References

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